

# Xinkai Shu

## Contact Information

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|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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## Research Interests

Online algorithms  
Approximation algorithms  
Shortest path algorithms

## Academic Positions

|              |                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------|
| 2026–present | <b>Professor (Tenure-Track)</b><br>School of Computer Science and Technology<br>University of Science and Technology of China |
| 2024–2026    | <b>Postdoctoral Researcher</b><br>Department 1: Algorithms and Complexity<br>Max Planck Institute for Informatics             |

## Education

|           |                                                                                                               |                         |
|-----------|---------------------------------------------------------------------------------------------------------------|-------------------------|
| 2019–2024 | <b>Doctor of Philosophy</b><br>Computer Science<br><i>The University of Hong Kong</i>                         | Supervisor: Zhiyi Huang |
| 2015–2019 | <b>Bachelor of Engineering</b><br>Computer Science and Technology (“Yao Class”)<br><i>Tsinghua University</i> |                         |

## Conference Publications

Authors are listed in alphabetical order following the traditions of theoretical computer science, unless stated otherwise.

- [1] A Faster Directed Single-Source Shortest Path Algorithm [[arXiv](#)] [[DOI](#)]  
Ran Duan, Xiao Mao, Xinkai Shu, and Longhui Yin  
*53rd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2026)*
- [2] The Long Arm of Nashian Allocation in Online p-Mean Welfare Maximization [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang, Chui Shan Lee, Xinkai Shu, and Zhaozi Wang  
*52nd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2025)*

- [3] Breaking the Sorting Barrier for Directed Single-Source Shortest Paths [[arXiv](#)] [[DOI](#)]  
Ran Duan, Jiayi Mao, Xiao Mao, [Xinkai Shu](#), and Longhui Yin  
*57th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2025)*  
**Best Paper Award. Accepted to the Journal of the ACM**
- [4] Online Matching Meets Sampling Without Replacement [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang, Chui Shan Lee, Jianqiao Lu, and [Xinkai Shu](#)  
*20th Conference on Web and Internet Economics (WINE 2024)*
- [5] Online Nash Welfare Maximization Without Predictions [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang, Minming Li, [Xinkai Shu](#), and Tianze Wei  
*19th Conference on Web and Internet Economics (WINE 2023)*
- [6] A Randomized Algorithm for Single-Source Shortest Path on Undirected Real-Weighted Graphs [[arXiv](#)] [[DOI](#)]  
Ran Duan, Jiayi Mao, [Xinkai Shu](#), and Longhui Yin  
*64th IEEE Annual Symposium on Foundations of Computer Science (FOCS 2023)*
- [7] The Power of Multiple Choices in Online Stochastic Matching [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang, [Xinkai Shu](#), and Shuyi Yan  
*54th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2022)*
- [8] Online Stochastic Matching, Poisson Arrivals, and the Natural Linear Program [[arXiv](#)] [[DOI](#)]  
Zhiyi Huang and [Xinkai Shu](#)  
*53rd Annual ACM SIGACT Symposium on Theory of Computing (STOC 2021)*

## Honors and Awards

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| 2025        | <b>Best Paper Award</b><br><i>57th Annual ACM SIGACT Symposium on Theory of Computing</i> |
| 2019        | <b>Postgraduate Scholarship</b><br><i>The University of Hong Kong</i>                     |
| 2016        | <b>Tsinghua-Baidu Scholarship</b><br><i>Tsinghua University &amp; Baidu, Inc.</i>         |
| 2014 & 2013 | <b>Gold Medal</b><br><i>China National Olympiad in Informatics</i>                        |

## Theses

Online Matching and Resource Allocation under Stochasticity  
*Doctoral thesis, supervised by Prof. Zhiyi Huang, 2024*

Quick Algorithms for Dynamic Edge Coloring (in Chinese)  
*Bachelor's thesis, Tsinghua University, 2019*

## Invited Talks

Breaking the Sorting Barrier for Directed Single-Source Shortest Paths  
– *International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW) 2025*  
Sorbonne University, June 30–July 2, 2025

A Randomized Algorithm for Single-Source Shortest Path on Undirected Real-Weighted Graphs

– CCF Forum for distinguished Ph.D. Candidates in Theoretical Computer Science 2023

The Hong Kong Polytechnic University, July 20–21, 2023

– Institute for Theoretical Computer Science (ITCS) Seminar

Shanghai University of Finance and Economics, July 11, 2023

Online Nash Welfare Maximization Without Predictions

– Complexity & Algorithms Workshop (C&A) 2023

Shandong University, April 1–2, 2023

The Power of Multiple Choices in Online Stochastic Matching

– International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW) 2022

City University of Hong Kong, August 15–19, 2022 (virtually)

## Research Visits

Mar–Aug 2023    Institute for Theoretical Computer Science  
Shanghai University of Finance and Economics

## Academic Services

Journal review:

- IEEE Transactions on Networking
- Acta Informatica

Conference review:

- 2027: SODA
- 2026: ESA, FOCS, STOC, SODA
- 2025: WAOA, APPROX, ICML, STOC, SODA
- 2024: ICALP, SODA
- 2023: ICALP, ISSAC, SODA
- 2022: STOC

## Teaching Experience

Teaching assistant at *the University of Hong Kong*:

- |               |            |                                              |
|---------------|------------|----------------------------------------------|
| – 2022 Spring | COMP 3250B | Design and analysis of algorithms            |
| – 2021 Fall   | COMP 3250A | Design and analysis of algorithms (Advanced) |
| – 2021 Fall   | COMP 3351  | Advanced algorithm analysis                  |
| – 2020 Spring | COMP 3250A | Design and analysis of algorithms            |

## Additional Information

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|-------------|-----------------------------------------------------------------------------|
| Languages   | Chinese (native), English (fluent), Japanese (intermediate), German (basic) |
| Programming | C++, Python, Pascal; former competitive programmer                          |
| Interests   | Logical puzzle games; chess problems, Go problems, <i>tsume-shogi</i>       |